

Data Analytics Insights Report

Smart City Mobility Analytics Team — Westbound I-94, Minneapolis–St Paul

Prepared for: City government traffic management stakeholders

Scope: ~48,000 hourly traffic records (Oct 2012 – Sep 2018), combined with weather and US federal holiday data

Summary

Westbound I-94 traffic shows a strong, predictable daily commute rhythm rather than being driven by weather. Congestion (>5,500 vehicles/hour) occurs in roughly **15% of recorded hours**. Weather condition and temperature have only a **weak statistical relationship** with traffic volume (temperature–volume correlation $r \approx 0.14$; clear weather is associated with modestly *lower* odds of congestion than cloudy weather, odds ratio ≈ 0.74), meaning that time-of-day and day-of-week patterns, not weather, should be the primary lens for congestion management and travel-time advisories on this corridor.

Key Findings

1. Traffic has grown over time, but raw yearly totals are misleading. The dataset's hourly sensor coverage varies enormously by year (from ~2,100 recorded hours in 2012 to ~8,700 in 2017), so comparing raw annual sums overstates or understates real trends. Looking instead at **average volume per recorded hour**, 2017 shows the highest typical traffic of the six years compared, and the trend across fully-recorded years points upward, consistent with growing commuter demand on the corridor rather than declining usage.

Recommendation: any capacity-planning figures derived from this dataset should be normalised per recorded hour, not by year-total, and the city should investigate the sensor/data-collection gaps directly.

2. Traffic volume is highly variable, and its distribution is roughly symmetric but very spread out. Mean and median volume are close ($\approx 3,290$ vs $\approx 3,427$ vehicles/hour), but the standard deviation ($\approx 1,985$) is 60% of the mean, traffic swings from near-empty overnight to over 7,000 vehicles/hour at peak. A single “average volume” figure is not a reliable planning number on its own; it must be read alongside the hourly pattern.

3. Weather is a weak predictor of congestion. The temperature–traffic correlation is positive but weak ($r \approx 0.14$, $R^2 \approx 1.9\%$, temperature explains under 2% of the variation in volume). The probability analysis confirms this: congestion and clear weather are not perfectly independent ($P(\text{Congestion} \cap \text{Clear}) = 0.0435$ vs. an independence expectation of 0.0496), but the effect size is modest (odds ratio ≈ 0.74).

Recommendation: weather-triggered traffic alerts should be treated as a minor signal at most; time-of-day/day-of-week based systems will be far more effective (see Part 2/3 modelling).

4. Holiday traffic dips are a calendar effect, not a weather effect. New Year's Day (near-freezing/sub-zero) and Labor Day (mild/warm) sit at opposite ends of the annual temperature cycle, as expected. Because both are public holidays, the traffic reduction on these dates is driven by fewer work trips, not by temperature. One data-quality note: **New Year's Day 2015 has zero recorded hours** (a genuine sensor/API gap), so it had to be excluded from that year's comparison.

5. The Power BI dashboard confirms these patterns with concrete numbers. Across all ~48,204 hourly records, average traffic volume is **3,260 vehicles/hour** (close to, though not identical to, the 3,290 de-duplicated figure in Finding 2, the dashboard uses the full raw dataset) and average temperature is **8.06°C**.

The 2017 hourly chart shows the expected two-peak commute shape: a **morning peak at 7am** ($\approx 4,880$ vehicles/hour) and a larger **evening peak at 4pm** ($\approx 5,800$ vehicles/hour, $\sim 19\%$ higher), larger because this is one westbound direction of the corridor, not a symmetric two-way count. By weather condition, **Clouds** has the highest average volume ($\approx 3,618$) and **Squall** the lowest ($\approx 2,062$), a gap of $\approx 1,557$ vehicles/hour, best read as squalls being rare winter events that coincide with fewer trips, not a condition that itself suppresses demand. The temperature-vs-volume scatter (after excluding one known 0 Kelvin sensor error, fixed the same way as in Part 2) shows a shallow trendline of about **+21 vehicles/hour per °C**, only $\sim 1,450$ vehicles/hour of swing across the entire -30°C to $+37^{\circ}\text{C}$ range observed, small next to the $\sim 2,000$ -vehicle overnight-to-peak swing seen every day. High-traffic ($>5,500$) readings cluster between **1°C and 19°C** (median 11°C), comfortable, non-extreme temperatures, reinforcing that weather modestly shapes discretionary travel rather than driving commuting itself.

Implications for the Mobility Team

- **Prioritise time-based interventions** (signal timing, congestion pricing windows, public-transit scheduling) over weather-based ones, the data does not support weather as a major lever.
- **Treat 2012–2016 year-total comparisons with caution** in any downstream reporting; use average-per-hour or fully-recorded-year comparisons instead, and flag the sensor-coverage gaps to whoever owns the data pipeline.
- **Use holiday-calendar effects, not seasonal-temperature effects**, to plan for traffic dips around public holidays.
- These findings carry into Part 2 (a reproducible Python pipeline that formalises the cleaning and feature engineering implied here) and Part 3 (predictive models that use time-based features, since they matter far more than weather, to forecast congestion and recommend travel windows).